

Supplementary Material for FiHARD

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1 Theoretical Properties of the Spatio-Spectral Forward Process

Proposition 1 (Validity of the spatio-spectral forward process). *Let*

$$\Sigma_{\text{space}} = U \Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_F),$$

with U orthonormal and $\lambda_m \geq 0$, and let $\nu_\ell \geq 0$ denote the frequency weights associated with the future DCT modes. Define the joint mode energies

$$\kappa_{\ell,m} = \nu_\ell \lambda_m,$$

and assume that the cumulative retention factors satisfy $0 < \bar{\alpha}_{t,\ell,m} \leq 1$ for every timestep t and every spatio-spectral mode (ℓ, m) . Then the forward process

$$z_t = \sqrt{\bar{\alpha}_t} \odot z_0 + \sqrt{1 - \bar{\alpha}_t} \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

with $z_0 = r_0 U$, is:

- 1. admissible, since each mode has nonnegative variance $1 - \bar{\alpha}_{t,\ell,m}$;*
- 2. linear-Gaussian in the residual coordinates, because it is diagonal Gaussian in z -space and mapped back by the orthonormal transform $r = zU^\top$;*
- 3. Markov, with one-step coefficients $\alpha_{t,\ell,m} = \bar{\alpha}_{t,\ell,m} / \bar{\alpha}_{t-1,\ell,m}$ defined mode by mode; and*
- 4. compatible with v -prediction, since together with z_t , the target*

$$v_t = \sqrt{\bar{\alpha}_t} \odot \epsilon - \sqrt{1 - \bar{\alpha}_t} \odot z_0$$

uniquely determines both z_0 and ϵ mode by mode.

2 Proof of Proposition 1

We prove each statement in turn.

Admissibility. By assumption, every cumulative retention factor satisfies

$$0 < \bar{\alpha}_{t,\ell,m} \leq 1.$$

Therefore each mode of the forward process has variance

$$1 - \bar{\alpha}_{t,\ell,m} \geq 0,$$

so the noising rule

$$z_{t,\ell,m} = \sqrt{\bar{\alpha}_{t,\ell,m}} z_{0,\ell,m} + \sqrt{1 - \bar{\alpha}_{t,\ell,m}} \epsilon_{\ell,m}$$

defines a valid Gaussian perturbation for every spatio-spectral mode.

Linear-Gaussian structure. In the spectral coordinates, the process is diagonal across modes and Gaussian because $\epsilon \sim \mathcal{N}(0, I)$ and the update is affine in (z_0, ϵ) . The residual coordinates are obtained through the linear map

$$r_t = z_t U^\top,$$

with U orthonormal. Linear images of Gaussian variables remain Gaussian, hence the induced process in residual coordinates is also linear-Gaussian.

Markov property. Let

$$\alpha_{t,\ell,m} = \frac{\bar{\alpha}_{t,\ell,m}}{\bar{\alpha}_{t-1,\ell,m}}.$$

Then the cumulative coefficients factorize exactly as in standard diffusion, mode by mode. Consequently, each coordinate admits a one-step transition of the form

$$z_{t,\ell,m} = \sqrt{\alpha_{t,\ell,m}} z_{t-1,\ell,m} + \sqrt{1 - \alpha_{t,\ell,m}} \eta_{t,\ell,m}, \quad \eta_{t,\ell,m} \sim \mathcal{N}(0, 1),$$

which depends on the past only through $z_{t-1,\ell,m}$. Since this holds independently for every mode, the full process is Markov.

Compatibility with v -prediction. For each mode, define

$$A_{t,\ell,m} = \sqrt{\bar{\alpha}_{t,\ell,m}}, \quad B_{t,\ell,m} = \sqrt{1 - \bar{\alpha}_{t,\ell,m}}.$$

The forward process and the v -target are

$$z_{t,\ell,m} = A_{t,\ell,m} z_{0,\ell,m} + B_{t,\ell,m} \epsilon_{\ell,m},$$

$$v_{t,\ell,m} = A_{t,\ell,m} \epsilon_{\ell,m} - B_{t,\ell,m} z_{0,\ell,m}.$$

This is a 2×2 linear system in the unknowns $(z_{0,\ell,m}, \epsilon_{\ell,m})$. Its coefficient matrix

$$\begin{bmatrix} A_{t,\ell,m} & B_{t,\ell,m} \\ -B_{t,\ell,m} & A_{t,\ell,m} \end{bmatrix}$$

has determinant

$$A_{t,\ell,m}^2 + B_{t,\ell,m}^2 = \bar{\alpha}_{t,\ell,m} + (1 - \bar{\alpha}_{t,\ell,m}) = 1.$$

Hence it is invertible, so $(z_{0,\ell,m}, \epsilon_{\ell,m})$ is uniquely determined by $(z_{t,\ell,m}, v_{t,\ell,m})$ for every mode. Therefore the proposed forward process is compatible with standard v -prediction.